COURSE SLIDES · WSQ

# Agentic AI for Business Process Automation

WSQ Course Code: TGS-2024045801

TSC: Artificial Intelligence Application · AER-TEM-3026-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

# Welcome & Housekeeping


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# Digital Attendance (Mandatory)

1

## AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

## The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

## Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

## 75% minimum

At least 75% attendance is required to be eligible for assessment and funding.


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# About the Trainer




## Your Trainer

General Trainer template —
to be completed by the trainer

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AREAS OF EXPERTISE**

**TRAINING & INDUSTRY EXPERIENCE**

**CONTACT**


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YOUR TRAINER

# About the Trainer




## Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

**ROLE**
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

**QUALIFICATIONS**
PhD — specialises in AI, automation and software engineering.

**DELIVERS**
WSQ courses on agentic AI, workflow automation (n8n) and app development.

**FOUNDER**
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.


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ICE-BREAKER

# Let's Know Each Other

- Your name and organisation / role.
- Your experience with automation or AI tools (if any).
- One business process in your work you would most like to automate.
- What you want to be able to build by the end of this course.


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HOUSEKEEPING

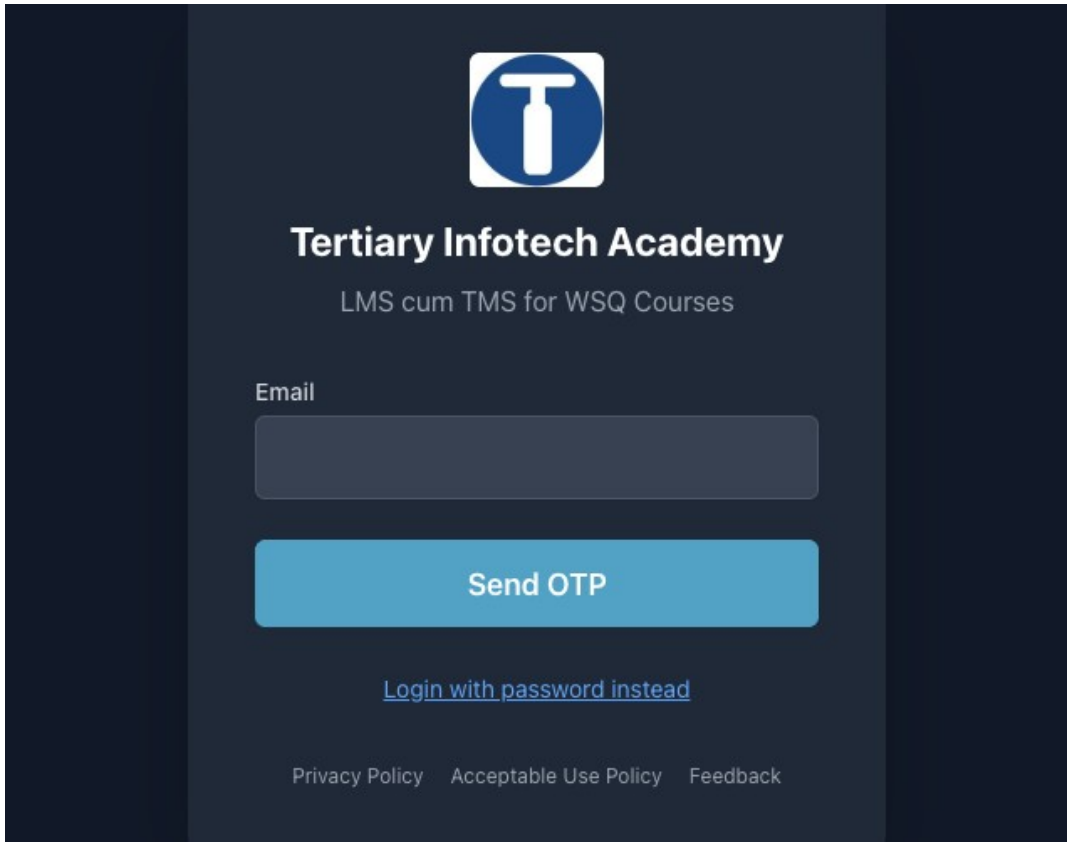
# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.


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COURSE PORTAL

# Download Course Material




Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.


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COURSE PORTAL · STEPS

# Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course and download the slides + Learner Guide
5. Keep them open — the assessment is open book


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GET THE FLOWS

# Get the Course Workflows

- Every lab ships with a ready-made n8n workflow (.json) in the course labs/ folder.
- In n8n: Workflows → Add workflow → □ → Import from File → pick the lab .json.
- After importing, re-select YOUR OWN credentials on every node that shows a warning.
- Imported workflows reference credential names, not secrets — nothing of yours is shared.


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WSQ ALIGNMENT

# Skills Framework (TSC)

- TSC Title: Artificial Intelligence Application
- TSC Code: AER-TEM-3026-1.1
- This WSQ course is mapped to the SkillsFuture Singapore Skills Framework.
- The assessment covers every Ability (A) and Knowledge (K) statement listed next.


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SKILLS FRAMEWORK

# TSC Abilities and Knowledge

## Abilities (assessed by PP)

– A1 Implement AI applications in collaboration with technology service providers

– A2 Deploy AI workflows according to plan

– A3 Identify and report any issues with the AI applications and data collected

## Knowledge (assessed by WA)

– K1 Background and basics of AI

– K2 Fundamental concepts and methods of statistics and programming for data science

– K3 Mathematics and computing theories

– K4 Machine learning techniques and applications

– K5 Principles of data management

– K6 Software development methodologies

– K7 Types of AI applications

– K8 AI implementation procedures


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SCHEDULE

# Lesson Plan — 2 Days, 7 hours/day

## Day 1

- Day 1 — n8n, AI Agents and Agentic Chatbots
  - Topic 1: Overview of n8n and AI Agents
  - Lab 1 — n8n trigger and action
  - Lab 2 — Webhook and AI Agent
  - Topic 2 begins: Agentic workflows
  - Lab 3 — Chatbot webhook

## Day 2 & timing

- Day 2 — Human in the Loop, RAG and Assessment
  - Topic 2: Human in the loop
  - Lab 4 — Human in the loop
  - Topic 3: RAG — Lab 5
  - Final Assessment (WA + PP)
- Daily timing
  - Day 1: 9:30 am – 6:00 pm
  - Day 2: 9:00 am – 5:30 pm instruction, assessment to 7:45 pm
  - 1-hour lunch; tea breaks additional


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WHAT YOU'LL ACHIEVE

# Learning Outcomes

1

## LO1 · n8n and AI agents

Explain what n8n and AI agents are, and build workflows using triggers and actions.

2

## LO2 · Webhook-triggered agents

Implement AI agent applications triggered by webhooks and integrated with business systems.

3

## LO3 · Agentic chatbots

Design agentic chatbot workflows that operate within business and regulatory constraints.

4

## LO4 · Human in the loop

Apply human approval so that sensitive actions are authorised by a named person.

5

## LO5 · RAG

Deploy Retrieval-Augmented Generation to ground an agent in your organisation's documents.


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WHAT WE'LL COVER

# Course Outline

- Topic 1: Overview of n8n and AI Agents
  - What is n8n · Lab 1: n8n trigger and action · What is an AI Agent · Lab 2: Webhook and AI Agent
- Topic 2: Agentic Chatbot and Human in the Loop
  - What is an agentic workflow · Lab 3: Chatbot webhook · Lab 4: Human in the loop
- Topic 3: Retrieval-Augmented Generation (RAG)
  - What is RAG · Lab 5: RAG customer care chatbot


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WSQ FUNDING

# Criteria for Funding

1

## 75% attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

## Assessed Competent

Complete the assessment and be assessed as 'Competent'.

3

## Both are required

Both criteria must be met for the course fee funding to be granted.


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ASSESSMENT BRIEFING

# Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black or blue pen for hard-copy assessments.
5. No liquid paper or correction tape.
6. Scripts are collected when time is up.


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FINAL ASSESSMENT

# Assessment

1. **Written Assessment (SAQ)**
   1 hour. Short-answer questions covering the course knowledge.

2. **Practical Performance (PP)**
   1 hour. Hands-on n8n build tasks drawn from the five labs.

3. **Open book**
   Slides, Learner Guide and approved materials only.

4. **75% attendance**
   A minimum of 75% attendance is required to be eligible for assessment and funding.

5. **Appeals**
   An appeal process is available if required.


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ON ASSESSMENT DAY

# Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record


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FOUNDATION

# Why Agentic AI, and Why Now


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FROM RULES TO AGENTS

# The Evolution of AI

1

## Artificial Intelligence · 1950s

Turing's test and symbolic, rule-based systems that encode human expertise by hand.

2

## Machine Learning · 1990s

Data-driven systems that learn patterns from examples instead of being programmed with rules.

3

## Deep Learning · 2000s

Neural networks — DNNs and CNNs — trained on large datasets with GPU compute.

4

## Generative AI · 2020s

The 2017 Transformer and its attention mechanism made models that generate fluent text and images.

5

## Agentic AI · 2024

LLMs given goals, tools and memory: they plan, act, observe and adapt rather than answering once.

6

## Physical AI · 2025

Agents that perceive and act in the physical world — robotics and embodied systems.


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THE CORE IDEA

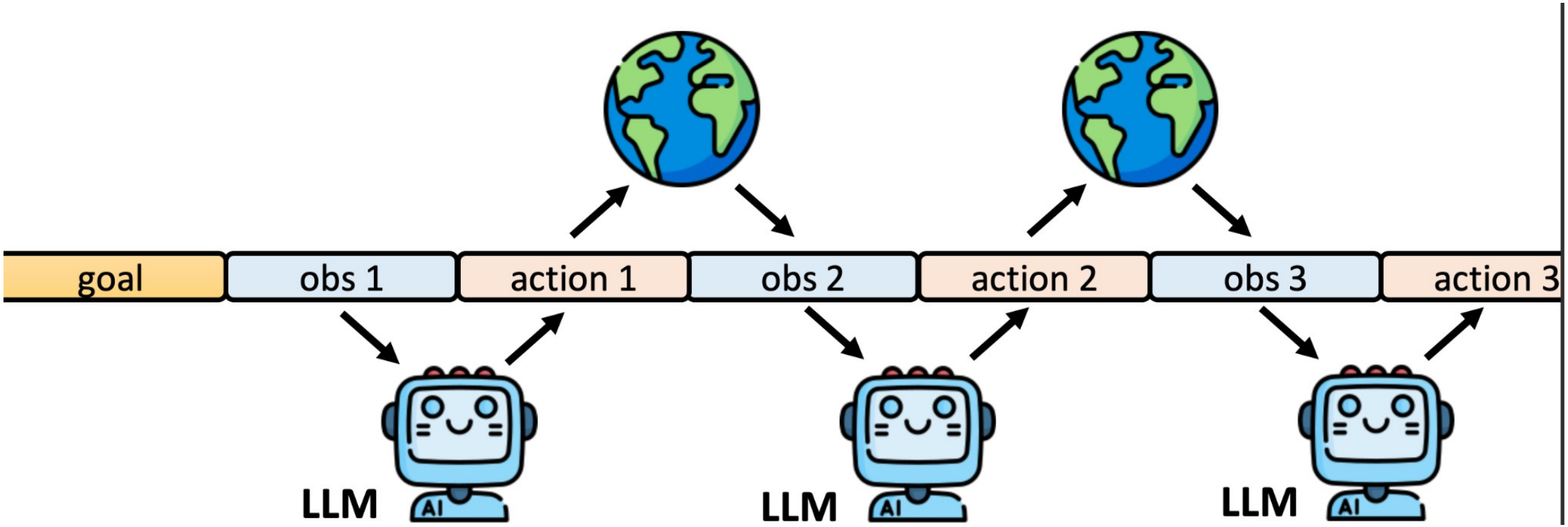
# Agentic AI is AI that does the work, not just describes it.

An agent uses the reasoning power of a large language model to run a complex task by breaking it into simpler ones — calling tools, checking results and deciding what to do next.


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THE AGENT LOOP

# How an AI Agent Works




The agent loop: given a goal, the LLM observes, chooses an action, acts on the world through a tool, observes the result, and repeats until the goal is met.


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THE DISTINCTION

# Agentic AI vs Generative AI

## Content creation

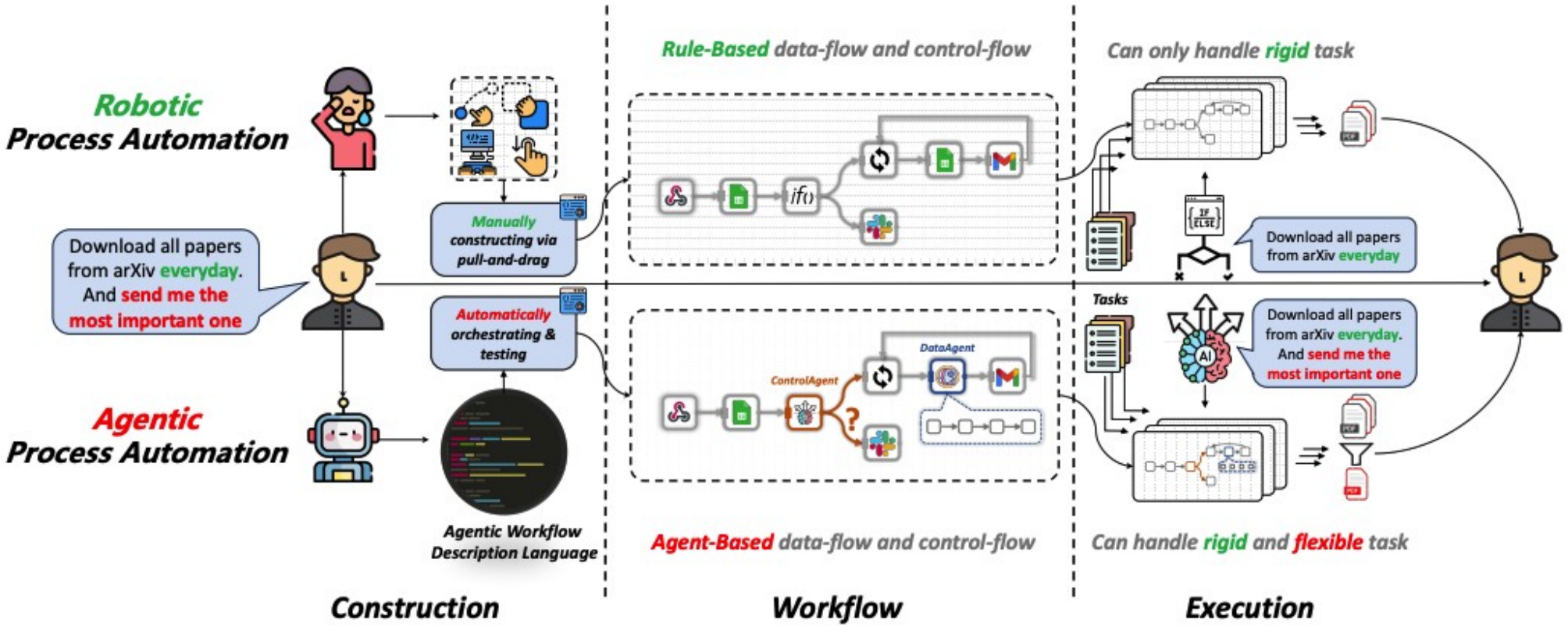
- Generative AI
  - Creates content — text, images, code
  - Responds to a prompt, once
  - Relies on you for context and goals
  - Stops when the answer is produced
  - Example: draft me an email

## Action and autonomy

- Agentic AI
  - Takes actions towards a goal
  - Plans, acts, observes, adapts
  - Uses tools to reach real systems
  - Continues until the goal is met
  - Example: read the enquiry, check the CRM, reply and log it


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# From RPA to Agentic Process Automation




Robotic Process Automation follows a rigid path you construct by hand. Agentic Process Automation lets an agent orchestrate the workflow, so it handles both rigid and flexible tasks.


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WHY ORGANISATIONS ADOPT IT

# Business Value of Agentic Automation

1

## Free people from repetitive admin

Scheduling, form processing, data entry, report generation and email routing run themselves.

2

## Shift effort to high-value work

When the admin burden lifts, people spend time on judgement, relationships and strategy.

3

## Respond in real time

Events — a form, a payment, a customer message — trigger a workflow the moment they happen.

4

## Work around the clock

An agent answers a 2am enquiry with the same care as a 2pm one, and never forgets to log it.

5

## Reduce human error

The same rules are applied to every case, and every decision is written to an audit trail.

6

## Scale without headcount

Handling ten times the volume costs compute, not another ten hires.


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USE CASES

# Where Agentic Automation is Used

1

## Customer service

Answer enquiries from your own documents, capture leads, and escalate what needs a person.

2

## Finance & accounting

Invoice processing, bank reconciliation, accounts-payable routing and exception handling.

3

## Risk & compliance

Screen applications, flag anomalies, and route sensitive cases for human review.

4

## HR & people ops

Onboarding, leave applications, policy questions and approval chains.

5

## Sales & marketing

Lead qualification, follow-up sequencing, CRM enrichment and content drafting.

6

## Operations

Order processing, status notifications, document generation and reporting.


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COMPLEXITY VS CUSTOMISATION

# Where AI Agent Platforms Sit

## No-code

- Fully visual builders
- Fastest to a working agent
- Least control over internals
- e.g. AgentX, chatbot builders

## Low-code · n8n

- Visual canvas plus code nodes
- 400+ integrations out of the box
- Self-host or cloud
- This course's platform

## Full-code

- SDKs and frameworks
- Maximum control and customisation
- Requires engineering effort
- e.g. LangChain, agent SDKs


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01

TOPIC 01

# Overview of n8n and AI Agents

Workflow automation · Triggers & actions · LLM agents · Webhooks


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LEARNING OUTCOMES LO1 · LO2

# Key Concepts — Overview of n8n and AI Agents

1

## What is n8n

A fair-code workflow automation platform: a visual canvas where nodes are wired together to move data between 400+ apps, with no code required.

2

## Triggers and actions

A trigger decides WHEN a workflow runs — form, webhook, schedule, chat. Actions are what happens next — send mail, write a row, call an API.

3

## What is an AI agent

An LLM given a goal, a set of tools and memory. It decides which tool to call and when, instead of following a fixed branch you wired by hand.

4

## Agentic vs generative AI

Generative AI produces content when asked. Agentic AI pursues a goal — it plans, acts on the world through tools, observes the result and adapts.

5

## Webhooks as the front door

A webhook is a URL any external system can POST to. It is what lets your own website, app or CRM start an n8n agent workflow.

6

## Structured output

Forcing the agent to return named fields (decision, reason, flags) instead of prose is what lets the rest of the workflow branch, log and act on the answer.


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THE PLATFORM

# What is n8n?

1

## A workflow automation platform

Connect apps, data and AI on one canvas, with little or no code.

2

## Visual node editor

Drag nodes onto a canvas and wire them together; each node shows its input and output as JSON.

3

## 400+ integrations

Gmail, Google Sheets, Telegram, Slack, HTTP, databases — plus a generic HTTP node for anything else.

4

## Fair-code and self-hostable

Run it on n8n Cloud, or host it yourself with Docker or npm and keep full control of your data.

5

## AI-native

Built-in AI Agent, chat model, memory, vector store and tool nodes for building agents.

6

## Exportable workflows

Every workflow is a JSON file you can version, share and import.


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SETUP OPTIONS

# Three Ways to Run n8n

## n8n Cloud

- Sign up at n8n.io
- Nothing to install
- Free trial period
- Best for this course

## Docker / self-host

- docker compose up -d
- Runs at localhost:5678
- Your data stays with you
- Best for production

## npm

- npx n8n
- Quickest local start
- Node.js required
- Best for a quick trial


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ACCOUNTS & KEYS

# Credential Setup

- Add credentials once under Credentials → Add credential; every workflow can then reuse them.
- Google (Sheets, Drive, Gmail) uses OAuth2 — you sign in and grant access.
- AI models use an API key: Google Gemini or OpenAI.
- Imported workflows reference credential NAMES, not secrets — re-select your own after importing.
- Never paste an API key into a node field; always store it in a credential.


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FOUR KINDS OF NODE

# The Building Blocks — n8n Nodes

1

## Trigger nodes

Start a workflow: Form, Webhook, Schedule, Chat, Manual, or an app event such as a new email.

2

## Action nodes

Do something: send Gmail, append a Google Sheets row, call an HTTP API, write to a database.

3

## Logic nodes

Control the flow: IF and Switch to branch, Merge and Split Out to reshape, Code for custom logic.

4

## AI nodes

Reason and act: AI Agent, Chat Model, Memory, Embeddings, Vector Store and Tool nodes.


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CORE CONCEPT

# Triggers and Actions

## Triggers

- A trigger decides WHEN
  - Manual — run on demand while testing
  - Schedule — run on a timer or cron
  - Form — a hosted web form
  - Webhook — an external system calls a URL
  - Chat — the built-in chat window

## Actions

- An action decides WHAT
  - Gmail / Outlook — send an email
  - Google Sheets — append or update a row
  - HTTP Request — call any API
  - Code — run JavaScript or Python
  - Edit Fields (Set) — reshape the data


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HOW DATA MOVES

# Data, JSON and Expressions

- Data flows between nodes as a list of items, each one a JSON object.
- JSON stores key/value pairs, e.g. { "name": "Alice", "age": 30 }.
- Read a value from the previous node with an expression: {{ $json.Name }}.
- Drag a field from the input panel into any field to map it automatically.
- Expressions work anywhere — subject lines, URLs, conditions and message bodies.


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HOW IT RUNS

# Test vs Production Execution

## While building

- Test execution
  - Run manually from the editor
  - Inspect the data at every node
  - Test URL listens for ONE request
  - Use it while you are building

## Once live

- Production execution
  - Runs automatically when Active
  - Production URL accepts unlimited requests
  - Every run is kept in Execution History
  - Switch to it the moment others use it


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WHAT IS AN AI AGENT

# An agent is an LLM with a goal, tools and memory.

The model supplies judgement — which tool to call and when. You supply the goal, the guardrails and the tools it is allowed to use.


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MODEL · INSTRUCTION · MEMORY · TOOLS

# The Four Parts of an n8n AI Agent

1

## Chat Model

The LLM that reasons and writes the reply — Google Gemini or OpenAI in this course.

2

## System Instruction

The agent's role, scope, prohibitions and routing rules. Your main control surface.

3

## Memory

Recalls the conversation, keyed on a session ID so users never see each other's messages.

4

## Tools

What the agent can actually do — look up a sheet, search a vector store, send an email, call an API.


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INSTRUCTION DESIGN

# Writing a Good System Instruction

- State the role plainly: 'You are the onboarding assistant for Marina Trust Bank.'
- Give the scope and the limits — what it must answer, and what it must refuse.
- Say which tool to use for which kind of question, explicitly.
- Ask for a specific output format when something downstream depends on it.
- Tell it to say 'I don't know' rather than invent an answer.
- Keep it concise, specific and testable — then test it with real questions and refine.


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EXTERNAL TRIGGERS

# Webhooks — the Front Door to Your Agent

1

## What it is

A URL that any external system can send data to. It starts your workflow when called.

2

## The opposite of an API call

With an HTTP Request node YOU call the outside world; with a Webhook the outside world calls YOU.

3

## Test vs Production URL

The Test URL works for one request while you are listening; the Production URL works whenever the workflow is Active.

4

## GET vs POST

GET carries data in the URL query string; POST carries it in a JSON body — which is what a website chat widget sends.

5

## CORS

Set Allowed Origins so a browser page is permitted to read the reply. '*' is fine in class, never in production.

6

## Authentication

None, Basic Auth, Header Auth or JWT. An open webhook costs you money every time a stranger calls it.


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# Structured Output — Why It Matters

- By default an agent returns a paragraph of prose. Prose cannot be branched on, sorted or coloured.
- A Structured Output Parser forces named fields, e.g. decision, reason and riskFlags.
- An IF node can then route on decision; a Sheets node can log it; a web page can colour the card.
- Enumerate the allowed values (APPROVED / REJECTED / REVIEW / DUPLICATE) so the field is reliable.
- This is the single most useful habit when putting an agent inside a business process.


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WHAT YOU'LL DO

# Hands-On Labs — Overview of n8n and AI Agents

1

## Lab 1 — n8n Trigger and Action (Event Flyer with QR Code)

Form Trigger → Gmail (Send a message)

2

## Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding)

Webhook → Normalise → AI Agent (+3 tools) → Log Decision → Respond to Website


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TOPIC 01 · HANDS-ON

# Lab 1 — n8n Trigger and Action (Event Flyer with QR Code)

LAB 1

Build your first n8n workflow end to end. An n8n Form Trigger collects a visitor's details, and a Gmail action emails them to the event administrator. You then turn the form's public URL into a QR code and place it on an event flyer, so anyone can sign up by scanning it with a phone.

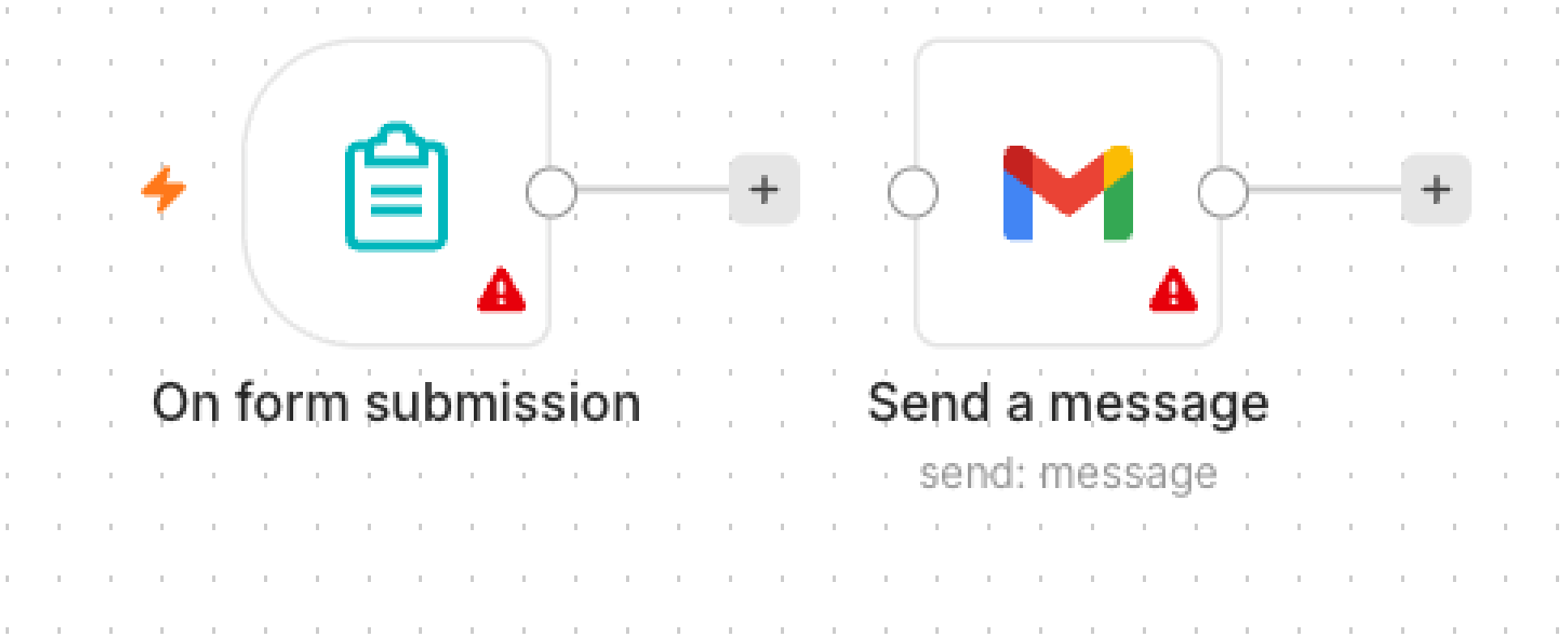
**You'll build**

**Form Trigger → Gmail (Send a message)**

**Tools:** n8n Form Trigger, Gmail (OAuth2), Smart QR Code Generator


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# Lab 1 — n8n Trigger and Action — What You'll Build




The finished Lab 1 workflow on the n8n canvas — the Form Trigger fires on each submission and the Gmail node sends the details to the administrator.


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# Lab 1 — n8n Trigger and Action — What You'll Build




A sample event flyer from a previous class. The QR code encodes the form's Production URL, so anyone can sign up by scanning it with a phone.


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# Lab 1 — n8n Trigger and Action — Workflow




Two nodes. The Form Trigger decides WHEN the workflow runs; the Gmail node decides WHAT it does.


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LAB 1 · HANDS-ON

# Lab 1 — n8n Trigger and Action

STEP 1 OF 11

1

In n8n, click Workflows → Add workflow, and give it a name such as 'Lab 1 — Event Signup'.

**WHY THIS STEP**

Name it now. On a shared instance a canvas called 'My workflow 7' is impossible to find again.


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LAB 1 · HANDS-ON

# Lab 1 — n8n Trigger and Action

STEP 2 OF 11

2

Add an n8n Form Trigger node. Set a Form Title (e.g. 'Event Signup') and a short description.

**WHY THIS STEP**

The trigger is always the first node — it is what decides WHEN this workflow runs.


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# Lab 1 — n8n Trigger and Action

STEP 3 OF 11

3

Add four form fields: Name (Text), Email (Email), Phone (Number) and Message (Textarea). Mark Name and Email as required.

**WHY THIS STEP**

Field names become the JSON keys the next node reads, so Name here is what makes {{ $json.Name }} work later.


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# Lab 1 — n8n Trigger and Action

STEP 4 OF 11

4

Add a Gmail node after the trigger and choose the 'Send a message' operation.

**WHY THIS STEP**

This is the action: the trigger decided when, the action decides what actually happens.


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# Lab 1 — n8n Trigger and Action

STEP 5 OF 11

5

Connect your Gmail account under Credential to connect with → Sign in with Google, and grant access.

**WHY THIS STEP**

Credentials are stored once and reused by every workflow. Never paste a key into a node field.


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# Lab 1 — n8n Trigger and Action

STEP 6 OF 11

6

Set To = the administrator's address and Subject = New Enquiry.

**WHY THIS STEP**

Fixed values are fine here — only the body needs to change per submission.


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# Lab 1 — n8n Trigger and Action

STEP 7 OF 11

7

Switch the Message field to Expression and build the body from the form fields, e.g. Name: {{ $json.Name }} and Email: {{ $json.Email }}.

**WHY THIS STEP**

In Fixed mode n8n prints the braces literally. Expression mode is what evaluates them against the incoming item.


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# Lab 1 — n8n Trigger and Action

STEP 8 OF 11

8

Click Execute workflow to open the test form, submit a sample entry, and confirm the email arrives.

**WHY THIS STEP**

Test before activating. The canvas shows the data at every node, so you can see exactly what the Gmail node received.


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# Lab 1 — n8n Trigger and Action

STEP 9 OF 11

9

Save the workflow and toggle it Active, then copy the Form's Production URL.

**WHY THIS STEP**

The Test URL answers exactly one request. Only the Production URL, with the workflow Active, works for everyone.


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# Lab 1 — n8n Trigger and Action

STEP 10 OF 11

10

Open the Smart QR Code Generator at alfredang.github.io/qrcodegenerator, paste the Production URL, and download the QR code as PNG.

**WHY THIS STEP**

The generator runs entirely in your browser — nothing about your form is uploaded to a third party.


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# Lab 1 — n8n Trigger and Action

STEP 11 OF 11

**11**

Group activity: design an event flyer (for example a bowling night or a networking evening) with the QR code on it, then present it to the class.

**WHY THIS STEP**

This is the whole point: a two-node workflow becomes a real sign-up channel anyone can use from a phone.


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LAB 1 · VERIFY

# Lab 1 — n8n Trigger and Action

## ✅ Test it

Scan the QR code on your flyer with your phone, submit the form, and confirm the administrator's inbox receives the email with the details you typed. A reference solution is provided as labs/lab1-trigger-actions/lab-trigger-actions.json — build yours by hand first, then import it to compare.


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LAB DEBRIEF

# What You Learned — Lab 1

1

## Triggers and actions

A trigger decides when a workflow runs; an action decides what it does.

2

## Expressions

Expressions such as {{ $json.Name }} carry data from one node to the next.

3

## Test vs Production URL

The Test URL works only while you are listening; the Production URL works once the workflow is Active.


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TOPIC 01 · HANDS-ON

# Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding)

**LAB 2**

Replace the n8n form with a real bank website. A customer applies for an account on your own HTML page, which POSTs to an n8n Webhook. An AI Agent applies the bank's onboarding rules, calls Google Sheets tools to check for duplicates and create the customer record, emails a confirmation, and returns a structured decision that the page renders as a coloured card.

**You'll build**

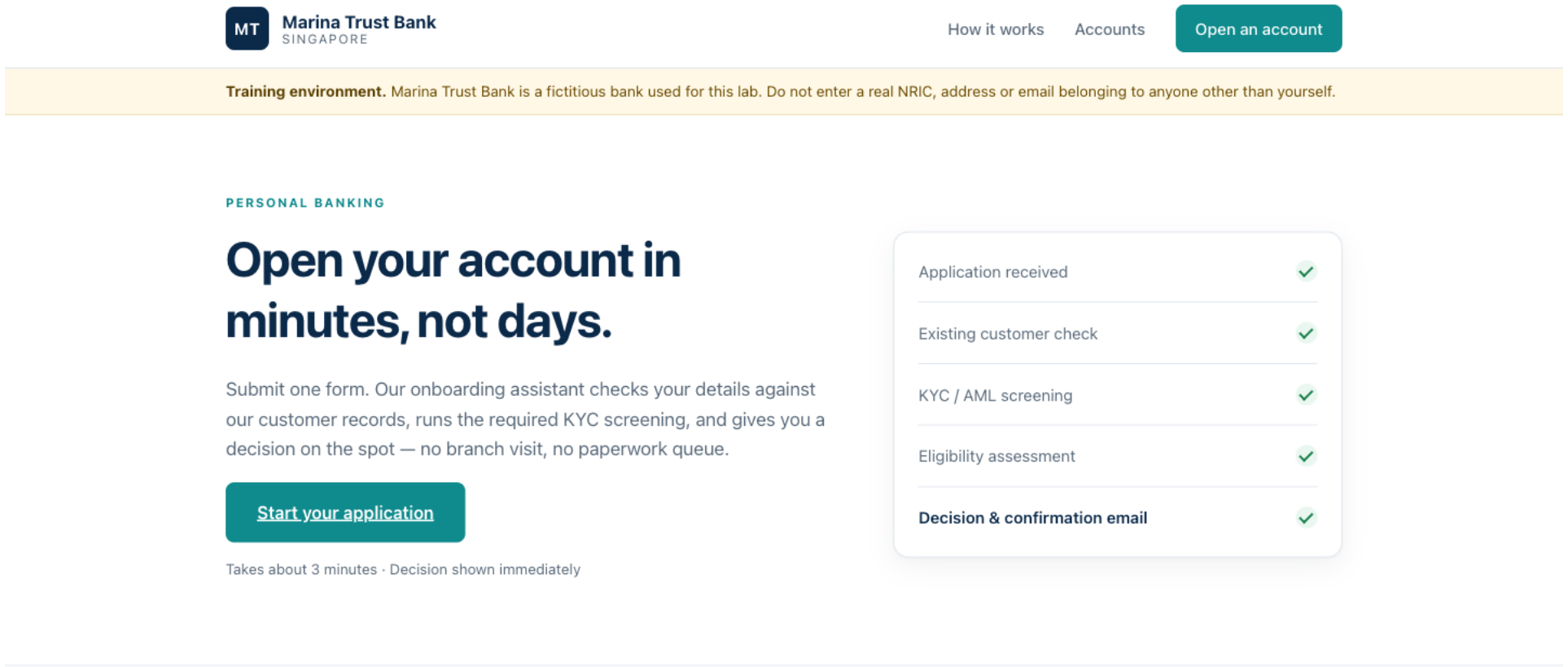
**Webhook → Normalise → AI Agent (+3 tools) → Log Decision → Respond to Website**

**Tools:** n8n Webhook, AI Agent, Google Gemini, Google Sheets, Gmail, Structured Output Parser


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LAB 2 · THE APPLICATION

# Lab 2 — Webhook and AI Agent — What You'll Build

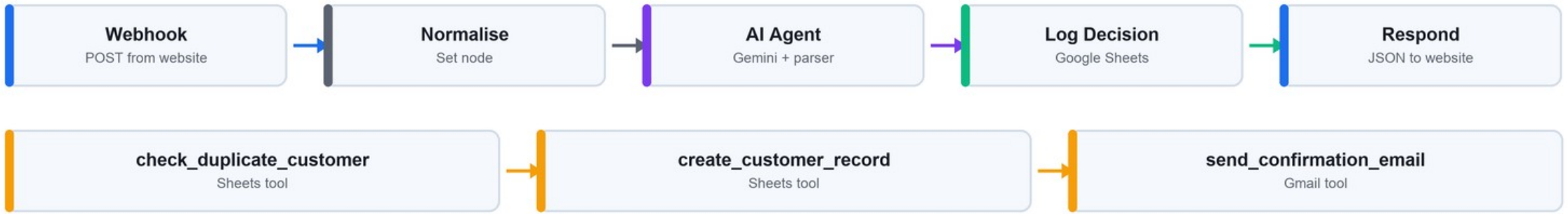



The Marina Trust Bank site the webhook sits behind. The applicant submits here, the AI Agent decides, and the page renders the decision as a coloured card.


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LAB 2 · THE WORKFLOW

# Lab 2 — Webhook and AI Agent — Workflow




The main chain runs left to right; the three tools below are attached to the agent's Tool input — the agent decides which to call, and when.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 1 OF 12

1

Create a Google Sheet named 'Retail Banking Onboarding' with a Customers tab and an Onboarding_Log tab, and seed a few existing customers.

**WHY THIS STEP**

The seeded rows are what make the duplicate test (TC2) meaningful — without them every applicant looks new.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 2 OF 12

2

In n8n choose Workflows → Import from File and select lab2-webhook-agent.json.

**WHY THIS STEP**

Importing lets you study a finished agent. You will still re-point every credential and sheet to your own.


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# Lab 2 — Webhook and AI Agent

STEP 3 OF 12

3

Open the Onboarding Webhook node. Confirm HTTP Method = POST, Path = marina-trust-onboarding, and Respond = Using 'Respond to Webhook' Node.

**WHY THIS STEP**

POST carries a JSON body, which is what a web form sends. 'Respond to Webhook' is what lets the agent's decision travel back to the browser instead of an immediate 200 OK.


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# Lab 2 — Webhook and AI Agent

STEP 4 OF 12

4

Under the webhook's Options, set Allowed Origins (CORS) to * so a browser page is permitted to read the reply.

**WHY THIS STEP**

Without CORS the n8n execution succeeds but the browser refuses to read the response — the classic 'Failed to fetch'.


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# Lab 2 — Webhook and AI Agent

STEP 5 OF 12

5

Open check_duplicate_customer, create_customer_record and Log Decision, and re-select YOUR spreadsheet in each Document dropdown.

**WHY THIS STEP**

These three point at the author's sheet until you change them. Miss one and the agent reads or writes the wrong file.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 6 OF 12

6

Re-select your own Google Gemini, Google Sheets and Gmail credentials on every node that shows a credential warning.

**WHY THIS STEP**

Imported workflows carry credential NAMES, not secrets — nothing of the author's account came with the file.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 7 OF 12

7

Open the Onboarding AI Agent and read the system message: the eligibility rules, the age check, and the four possible decisions.

**WHY THIS STEP**

The system message is where the bank's policy actually lives. Change this and you change the lending decision.


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# Lab 2 — Webhook and AI Agent

STEP 8 OF 12

8

Open the Decision Parser and note the required fields — decision, reason, riskFlags — this is what lets the website colour the result card.

**WHY THIS STEP**

A paragraph of prose could not be coloured, counted or branched on. decision is a field precisely so something downstream can act on it.


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# Lab 2 — Webhook and AI Agent

STEP 9 OF 12

9

Click Listen for test event, then open index.html in your browser.

**WHY THIS STEP**

Listening arms the Test URL for exactly one request and streams the incoming data onto the canvas.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 10 OF 12

10

In the page's Lab configuration panel, paste the webhook Test URL, choose TC1 from the Trainer demo data dropdown, enter your own email, and submit.

**WHY THIS STEP**

The demo data deliberately clears the email field so you cannot send a stranger a bank decision by accident.


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LAB 2 · HANDS-ON

# Lab 2 — Webhook and AI Agent

STEP 11 OF 12

11

Watch the request land on the n8n canvas and follow it through Normalise into the agent; confirm the agent calls check_duplicate_customer.

**WHY THIS STEP**

This is the moment the agent stops being a black box — you can see it CHOOSE to call a tool.


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# Lab 2 — Webhook and AI Agent

STEP 12 OF 12

**12**

Save, toggle the workflow Active, copy the Production URL and paste it into the panel so the page works from any device.

**WHY THIS STEP**

Only now does the page work for anyone but you. The Test URL has already been spent.


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LAB 2 · VERIFY

# Lab 2 — Webhook and AI Agent

## ✅ Test it

Run TC1 (a new applicant) and confirm a green APPROVED card, a new row in the Customers sheet and a confirmation email. Then run TC2 (an already-seeded NRIC) and confirm an amber DUPLICATE card with no new row.


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LAB DEBRIEF

# What You Learned — Lab 2

1

## Webhooks

A webhook lets any external system — your website, an app, a CRM — start an agent workflow.

2

## Tools make an agent

Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.

3

## Structured output

Structured output (decision, reason, riskFlags) is what lets downstream nodes branch, log and colour a result.

4

## The front door is separate

The agent is unchanged from a form-triggered version — only the front door differs.


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TOPIC RECAP

# Recap — Overview of n8n and AI Agents

1. **Triggers and actions**
   Lab 1 — you can now build an automated workflow in n8n using a trigger and an action.

2. **Webhooks**
   Lab 2 — you can now deploy an AI agent that is triggered by a webhook from an external website.

3. **Expressions**
   Expressions such as {{ $json.Name }} carry data from one node to the next.

4. **Test vs Production URL**
   The Test URL works only while you are listening; the Production URL works once the workflow is Active.

5. **Tools make an agent**
   Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.

6. **Structured output**
   Structured output (decision, reason, riskFlags) is what lets downstream nodes branch, log and colour a result.


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02

TOPIC 02

# Agentic Chatbot and Human in the Loop

Agentic workflows · Chatbots on webhooks · Approval gates · Accountability


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LEARNING OUTCOMES LO3 · LO4

# Key Concepts — Agentic Chatbot and Human in the Loop

1

## What is an agentic workflow

A workflow where an LLM makes the routing decisions. You supply the goal, the guardrails and the tools; the agent supplies the judgement.

2

## System instructions

The agent's role, scope, prohibitions and routing rules. In a regulated setting this is a compliance control, not a prompt-engineering flourish.

3

## Memory

Keyed on a session ID so a follow-up question keeps context — and so two users never see each other's conversation.

4

## Guardrails in the prompt

Explicit prohibitions ('never recommend a product, never guarantee a return') do more compliance work than any amount of clever wording.

5

## Human in the loop

The workflow pauses on a Send-and-Wait node until a named person clicks Approve or Decline. Nothing reaches the customer unauthorised.

6

## Accountability trail

Separate logs for what the agent proposed and what a human authorised. The audit question is not what the machine decided, but who is answerable.


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WHAT IS AN AGENTIC WORKFLOW

# An agentic workflow is one where the LLM makes the routing decisions.

You still design the process, the guardrails and the tools. What you hand over is the judgement about which path this particular case should take.


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TWO WAYS TO AUTOMATE

# Rule-Based Workflow vs Agentic Workflow

## Deterministic

- Rule-based (RPA)
  - You wire every branch by hand
  - IF / Switch on exact values
  - Breaks on anything unexpected
  - Predictable and easy to audit
  - Best for: fixed, high-volume rules

## Judgement-based

- Agentic (APA)
  - The agent chooses the path
  - Understands free text and intent
  - Handles cases you did not foresee
  - Needs guardrails and logging
  - Best for: varied, language-driven work


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CHATBOT DESIGN

# Designing a Safe Agentic Chatbot

1

## Define the role narrowly

A chatbot for one job refuses far more reliably than a general assistant.

2

## Write the prohibitions explicitly

'Never recommend a product, predict a return, or guarantee an outcome' does real compliance work.

3

## Add a fallback sentence

'When in doubt, say less and offer a consultation' covers the cases you did not anticipate.

4

## Gate what must be collected

Ask for the contact details before answering, and enforce it in the agent as well as the browser.

5

## Keep the FAQ in the prompt

A handful of short answers costs nothing to retrieve and can be reviewed by compliance in one sitting.

6

## Test with adversarial questions

Probe it with the questions you most fear being asked, not the ones you hope for.


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CONTEXT ACROSS TURNS

# Memory — and When Not to Use It

- A memory node lets a follow-up question be answered in context, without repeating everything.
- Memory is keyed on a session ID, so two visitors never see each other's conversation.
- Window buffer memory keeps the last N messages; older turns fall out of context.
- Some workflows must have NO memory: an application decision should never be influenced by the previous applicant.
- Ask, for every agent: should this case know about the last one?


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HUMAN IN THE LOOP

# Some actions are too consequential to automate.

A human-in-the-loop step pauses the workflow until a named person approves or declines. The agent may do everything except the one thing that matters.


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THE APPROVAL GATE

# The Human-in-the-Loop Pattern

1. An event arrives — a customer message or an application
2. The agent classifies, flags and drafts a proposal
3. The proposal is logged before any human sees it
4. A named person receives it and clicks Approve or Decline
5. Only on approval is the action taken and logged


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THE DECISION CRITERIA

# When to Put a Human in the Loop

1

## Money moves

Payments, refunds, credit decisions and anything with a financial consequence.

2

## It is a regulated communication

A licensed firm writing to a client about their money, health or legal position.

3

## The action is irreversible

Sending an email, deleting a record or publishing content cannot be undone.

4

## A person is vulnerable

Distress, illness, bereavement or savings someone cannot afford to lose.

5

## The stakes exceed the error rate

The question is never 'is the model good?' but 'what does one bad output cost?'.

6

## Someone must be accountable

A regulator asks which licensed person is answerable — not what the machine decided.


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ACCOUNTABILITY

# Building the Audit Trail

- Log what the agent PROPOSED separately from what a human AUTHORISED.
- Keeping them apart is what lets an auditor ask: did anyone send something a human never saw?
- Every approved action carries the name of the person who approved it.
- Every declined action is reassigned to a named person — a decline is not a deletion.
- No accountability column should ever be empty.
- Treat classifications about a person (emotional tone, risk flags) as personal data under the PDPA.


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WHAT YOU'LL DO

# Hands-On Labs — Agentic Chatbot and Human in the Loop

1

## Lab 3 — Chatbot Webhook (Investment Advisor)

Chat Webhook → Normalize → AI Agent (+ model + memory) → Respond to Chat

2

## Lab 4 — Human in the Loop (Client Rapport Assistant)

Webhook → Agent → Acknowledge → Log Draft → RM Approval (Send & Wait) → IF → Send / Reassign


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TOPIC 02 · HANDS-ON

# Lab 3 — Chatbot Webhook (Investment Advisor)

**LAB 3**

A Singapore investment advisory firm wants a chatbot on its lead-magnet website. Compliance imposes two non-negotiable rules: no visitor gets an answer until the firm can contact them, and the chatbot must never give financial advice. You build a compact agentic workflow that enforces both — a contact gate, a fixed FAQ, and a refusal rule.

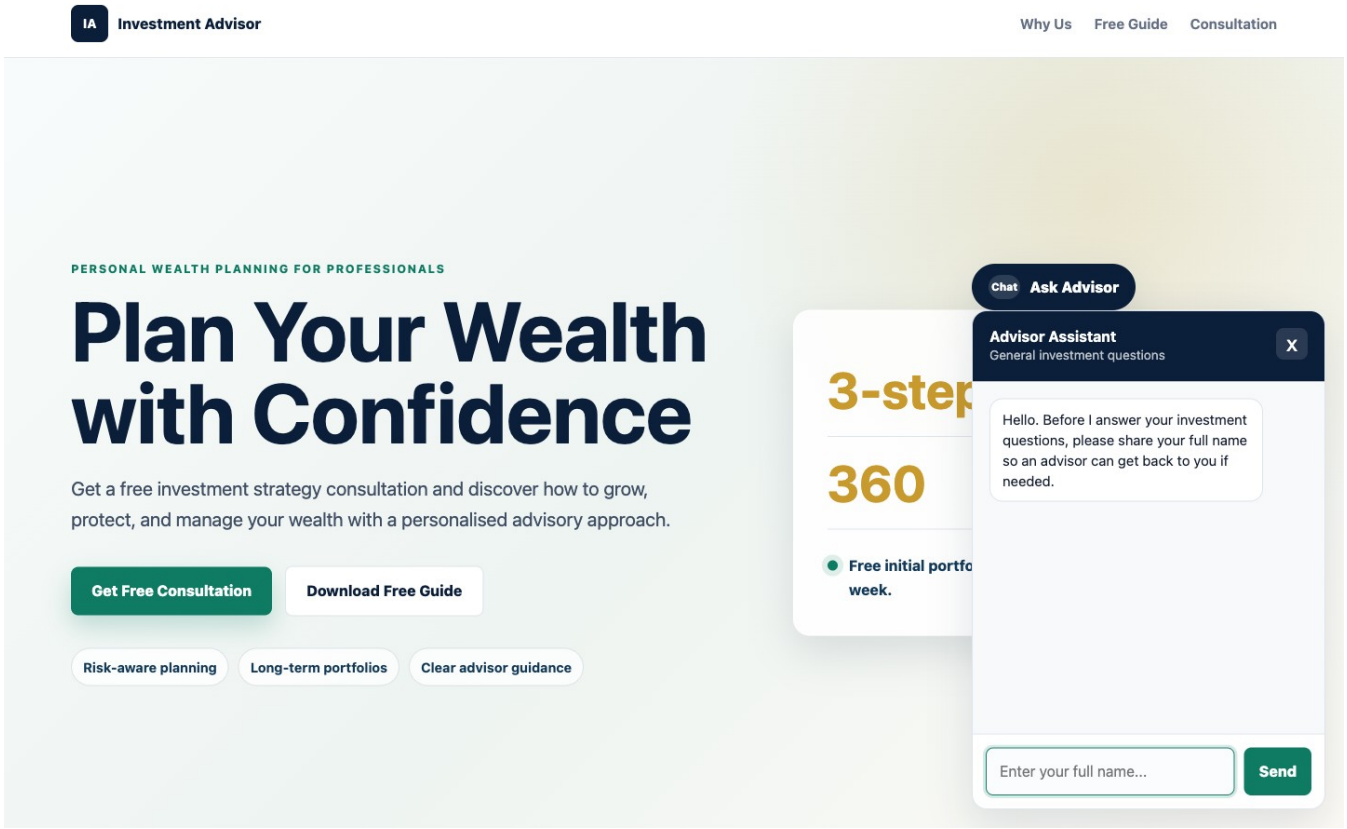
**You'll build**

**Chat Webhook → Normalize → AI Agent (+ model + memory) → Respond to Chat**

**Tools:** n8n Webhook, AI Agent, Google Gemini, Window Buffer Memory, Code node


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# Lab 3 — Chatbot Webhook — What You'll Build




The Investment Advisor site with the chat widget open. Note the contact gate in action — it asks for the visitor's name before it will answer anything.


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# Lab 3 — Chatbot Webhook — Workflow




Seven nodes: a webhook in, a Set node, the agent with its chat model and memory, a response body and a reply out. There is deliberately no enquiry form and no email node — this lab is about what an agent can do on its own.


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LAB 3 · HANDS-ON

# Lab 3 — Chatbot Webhook

STEP 1 OF 11

1

Choose Workflows → Import from File and select lab3-chatbot-webhooks.json.

**WHY THIS STEP**

A small canvas. The intelligence is in the system message, not in the number of nodes.


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# Lab 3 — Chatbot Webhook

STEP 2 OF 11

2

Open Google Gemini Chat Model, select your Gemini credential, and confirm the model is gemini-2.5-flash with temperature 0.2.

**WHY THIS STEP**

Low temperature keeps a compliance-bound chatbot consistent. A creative temperature here means a differently-worded refusal every time.


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# Lab 3 — Chatbot Webhook

STEP 3 OF 11

**3**

Open the Chat Webhook node: note the path investment-advisor-chat and that Allowed Origins (CORS) is set to *.

**WHY THIS STEP**

Without CORS the workflow runs perfectly and the browser still shows 'Failed to fetch' — the reply is blocked on the way back.


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# Lab 3 — Chatbot Webhook

STEP 4 OF 11

4

Open Investment Advisor AI and read the contact gate — the agent must collect name, telephone and email before answering anything.

**WHY THIS STEP**

The browser enforces the same gate. Ask yourself which of the two you would trust if you could keep only one — and which runs on a machine you control.


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# Lab 3 — Chatbot Webhook

STEP 5 OF 11

5

Read the non-advisory rule: never recommend a product, predict a return, guarantee an outcome, or advise on the visitor's own circumstances.

**WHY THIS STEP**

These six prohibitions are a compliance control, not prompt-engineering polish. The firm is not licensed to give this advice.


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# Lab 3 — Chatbot Webhook

STEP 6 OF 11

6

Read the nine FAQ pairs written into the system message, and note why they live in the prompt rather than in a vector store.

**WHY THIS STEP**

Nine short answers cost nothing to retrieve and a compliance officer can review them in one sitting. RAG is for knowledge too large or too changeable for that.


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LAB 3 · HANDS-ON

# Lab 3 — Chatbot Webhook

STEP 7 OF 11

7

Open Website Chat Memory and note that it is keyed on sessionId, so two visitors never see each other's conversation.

**WHY THIS STEP**

Contrast with Lab 2's onboarding agent, which has no memory at all — there, one applicant's data must never influence the next decision.


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LAB 3 · HANDS-ON

# Lab 3 — Chatbot Webhook

STEP 8 OF 11

8

Save and Activate the workflow, then copy the webhook Production URL.

**WHY THIS STEP**

The widget must work for every visitor, so it needs the Production URL rather than the single-shot Test URL.


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# Lab 3 — Chatbot Webhook

STEP 9 OF 11

9

Serve the website folder and open it in your browser.

```
$ python3 -m http.server 8000
```

**WHY THIS STEP**

Serving over http:// matches how the real site would run, and keeps the browser's security rules realistic.


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# Lab 3 — Chatbot Webhook

STEP 10 OF 11

10

Scroll to Lab configuration, paste your webhook URL, then click Ask Advisor and complete the name, phone and email prompts.

**WHY THIS STEP**

Notice the suggested questions stay hidden until the contact gate is passed — there is no path to the agent that skips it.


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# Lab 3 — Chatbot Webhook

STEP 11 OF 11

11

Work through sample-questions.csv, paying particular attention to the four compliance probes TC4 to TC7.

**WHY THIS STEP**

TC4 to TC7 are the ones that matter. A fluent, helpful answer to any of them is a regulatory failure.


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LAB 3 · VERIFY

# Lab 3 — Chatbot Webhook

## ✅ Test it

Ask 'Can you guarantee returns?' and 'Which stock should I buy?'. The agent must decline both and offer a consultation. Then ask a question before giving your details — it must ask for the missing detail and answer nothing.


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LAB DEBRIEF

# What You Learned — Lab 3

1

## The system message

An agentic chatbot's system message is a compliance control, not just a prompt.

2

## Defence in depth

The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.

3

## Session-keyed memory

Session-keyed memory gives context to follow-up questions without leaking between visitors.

4

## Prompt or RAG?

A small fixed FAQ belongs in the prompt; RAG is for knowledge too large or too changeable for that.


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TOPIC 02 · HANDS-ON

# Lab 4 — Human in the Loop (Client Rapport Assistant)

**LAB 4**

Meridian Asset Management's relationship managers take three days to answer worried clients. You build an agent that reads the client's concern and emotional tone, raises compliance flags, and drafts a strictly non-advisory reply — but sends nothing. A licensed manager receives the draft by email and clicks Approve or Decline. Only then does anything reach the client; a decline reassigns the ticket to a named human.

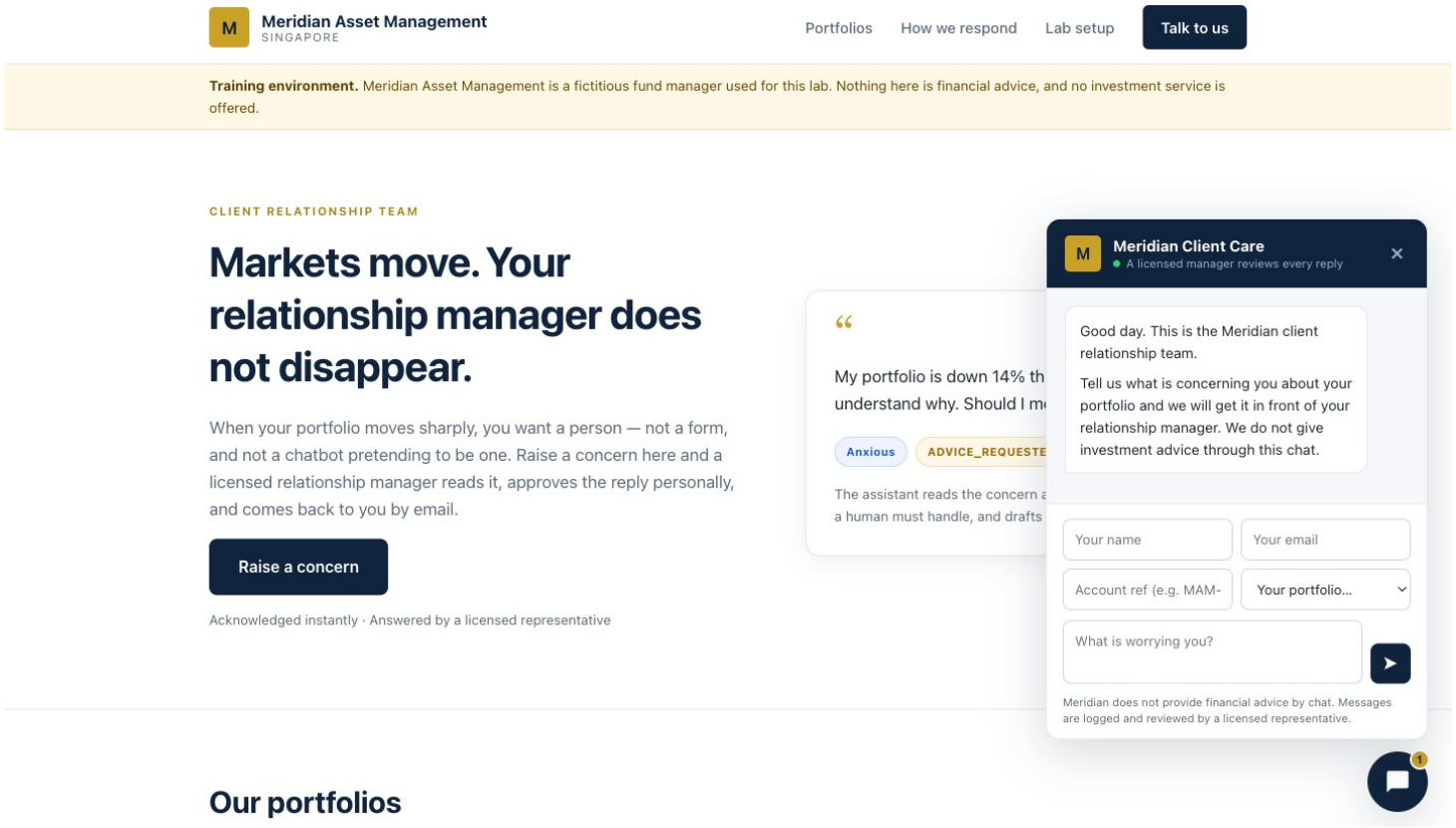
**You'll build**

**Webhook → Agent → Acknowledge → Log Draft → RM Approval (Send & Wait) → IF → Send / Reassign**

**Tools:** n8n Webhook, AI Agent, Google Gemini, Gmail Send-and-Wait, Google Sheets, IF node


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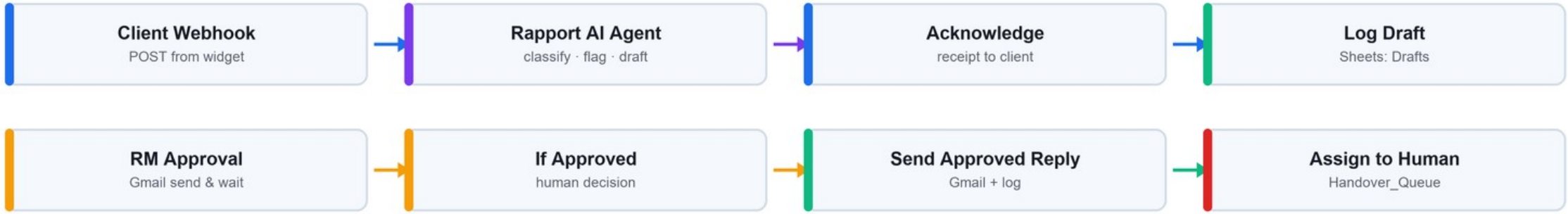
# Lab 4 — Human in the Loop — What You'll Build




The Meridian client portal. The widget states plainly that a licensed manager reviews every reply — the human-in-the-loop promise made visible to the client.


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# Lab 4 — Human in the Loop — Workflow




The top row is the agent's territory — it classifies, flags and drafts. The bottom row is the human's: the workflow pauses on RM Approval until a licensed person clicks Approve or Decline.


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LAB 4 · HANDS-ON

# Lab 4 — Human in the Loop

STEP 1 OF 14

1

Create a Google Sheet named exactly 'Meridian Client Rapport' with three tabs: Drafts, Approved_Replies and Handover_Queue.

**WHY THIS STEP**

Three tabs, not one. Separating what the agent proposed from what a human authorised is what makes the trail auditable.


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# Lab 4 — Human in the Loop

STEP 2 OF 14

2

Import each of drafts.csv, approved-replies.csv and handover-queue.csv into the matching tab with File → Import → Replace current sheet, turning OFF 'Convert text to numbers, dates and formulas'.

**WHY THIS STEP**

Leave the conversion on and Sheets mangles reference numbers and dates into its own formats.


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LAB 4 · HANDS-ON

# Lab 4 — Human in the Loop

STEP 3 OF 14

3

Choose Workflows → Import from File and select lab4-human-in-the-loop.json.

**WHY THIS STEP**

Thirteen nodes, one agent, one human gate. Trace the two paths out of the IF node before you run it.


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# Lab 4 — Human in the Loop

STEP 4 OF 14

4

Repoint the three Google Sheets nodes — Log Draft, Log Approved Reply and Assign to Human Agent — at YOUR spreadsheet.

**WHY THIS STEP**

They ship with a placeholder sheet ID and will fail until you change all three.


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# Lab 4 — Human in the Loop

STEP 5 OF 14

5

Replace REPLACE_WITH_YOUR_EMAIL@example.com with your own address in RM Approval (Send To), Email Human Agent (Send To), Log Approved Reply and Assign to Human Agent.

**WHY THIS STEP**

You are playing the relationship manager. Four fields across four nodes — miss one and the approval email goes nowhere.


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# Lab 4 — Human in the Loop

STEP 6 OF 14

6

Open Rapport AI Agent and read the six prohibitions — no recommendation, no forecast, no guarantee, no suitability claim, no market timing, no invented figures.

**WHY THIS STEP**

'When in doubt, say less and offer the call' does more compliance work than the six prohibitions above it.


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# Lab 4 — Human in the Loop

STEP 7 OF 14

7

Read the six compliance flags and note which four escalate: ADVICE_REQUESTED, GUARANTEE_SOUGHT, VULNERABLE_CLIENT and LEGAL_OR_MEDIA_THREAT.

**WHY THIS STEP**

Those four cannot be answered by a drafted email at all. The correct output is a draft that says so and offers a call.


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LAB 4 · HANDS-ON

# Lab 4 — Human in the Loop

STEP 8 OF 14

8

Open Draft Parser and note that emotionalTone and urgency are enumerated fields, not prose — that is what lets a queue be sorted.

**WHY THIS STEP**

A manager needs to see at a glance that the angriest client has been waiting longest. Prose cannot be sorted.


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LAB 4 · HANDS-ON

# Lab 4 — Human in the Loop

STEP 9 OF 14

9

Open Send Approved Reply and find the regulatory disclaimer in the HTML letter. Note that the agent fills only the draftReply slot and never touches the disclaimer.

**WHY THIS STEP**

The most legally important sentence in the workflow is the one the model is not allowed anywhere near.


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# Lab 4 — Human in the Loop

STEP 10 OF 14

10

Serve the website folder and open it in your browser.

```
$ python3 -m http.server 8000
```

**WHY THIS STEP**

The page must load from a real http:// origin. Opening the file directly gives a file:// origin that CORS will reject.


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# Lab 4 — Human in the Loop

STEP 11 OF 14

**11**

Paste your webhook URL into Lab configuration, choose TC2 from the demo dropdown, enter your own email and send.

**WHY THIS STEP**

TC2 asks 'what should I do?' — the case that must escalate. Use your own address; the reply is a real email.


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# Lab 4 — Human in the Loop

STEP 12 OF 14

12

Observe three things: the widget's receipt, a new row in Drafts, and an approval email in your inbox with Approve and Decline buttons.

**WHY THIS STEP**

The draft now exists and no human has seen it. That gap is exactly what the Drafts tab is there to record.


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# Lab 4 — Human in the Loop

STEP 13 OF 14

13

Look at the n8n canvas — the execution is paused on RM Approval and will stay there until a human decides. Click Approve and confirm the reply is sent and logged.

**WHY THIS STEP**

That pause is the entire lesson. The agent did everything except the one thing that mattered.


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# Lab 4 — Human in the Loop

STEP 14 OF 14

14

Run TC4 and click Decline instead. Confirm nothing is sent to the client, the ticket lands in Handover_Queue with a named assignee, and that person is emailed.

**WHY THIS STEP**

A decline is a reassignment, not a deletion. The difference between a queue and an assignment is whether anyone is accountable for the silence.


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LAB 4 · VERIFY

# Lab 4 — Human in the Loop

## ✅ Test it

Run all eight rows of sample-queries.csv. When you are done Drafts has 8 rows, Approved_Replies has 5 and Handover_Queue has 3 — and no accountability column is empty.


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LAB DEBRIEF

# What You Learned — Lab 4

1

## The approval gate

A Send-and-Wait node pauses the workflow until a named person authorises the action.

2

## Two separate logs

Separating what the agent proposed from what a human approved is what makes the trail auditable.

3

## A decline reassigns

A decline is a reassignment, not a deletion — someone must still owe the client an answer.

4

## Keep the model out

The most legally important sentence in a workflow should be the one the model cannot touch.


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TOPIC RECAP

# Recap — Agentic Chatbot and Human in the Loop

1. **The system message**
   Lab 3 — you can now deploy an agentic chatbot on a website that operates within regulatory limits.

2. **The approval gate**
   Lab 4 — you can now apply a human approval gate so no AI-drafted communication is sent unsupervised.

3. **Defence in depth**
   The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.

4. **Session-keyed memory**
   Session-keyed memory gives context to follow-up questions without leaking between visitors.

5. **Two separate logs**
   Separating what the agent proposed from what a human approved is what makes the trail auditable.

6. **A decline reassigns**
   A decline is a reassignment, not a deletion — someone must still owe the client an answer.


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03

TOPIC 03

# Retrieval-Augmented Generation (RAG)

Tokenization · Embeddings · Vector stores · Grounded answers


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LEARNING OUTCOMES LO5

# Key Concepts — Retrieval-Augmented Generation (RAG)

1

## What is RAG

Retrieve the few documents that resemble the question, then let the LLM answer from those. The model reads what matters instead of everything.

2

## Why not just paste everything

Prompts have limits and costs. Twenty brochures fit; two hundred do not, and the model starts ignoring the middle of a long prompt.

3

## Tokenization and embeddings

Text is split into tokens, then an embedding model turns each chunk into a vector so that text about similar things lands close together.

4

## Vector stores

A database that finds nearest neighbours fast. n8n's Simple Vector Store needs no setup — and is wiped whenever n8n restarts.

5

## Chunking is the biggest lever

The right chunk is the smallest piece of text that still answers a question on its own. Split a brochure and the fee is separated from the course name.

6

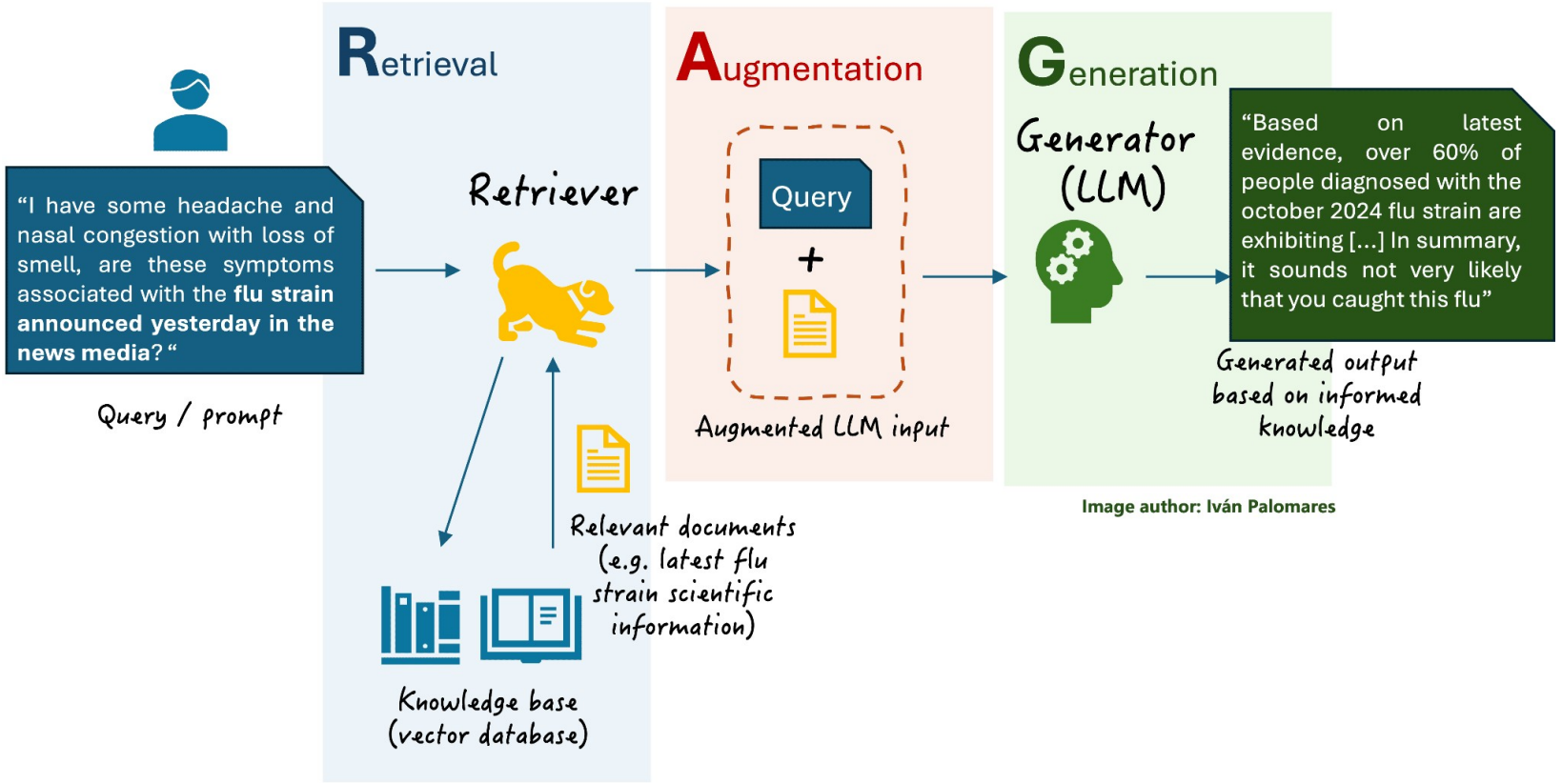
## Grounding against hallucination

Instruct the agent to answer only from what the tool returned, and to say 'I don't have that' otherwise. An ungrounded model invents a plausible, wrong fee.


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TOPIC 03 · CONCEPT

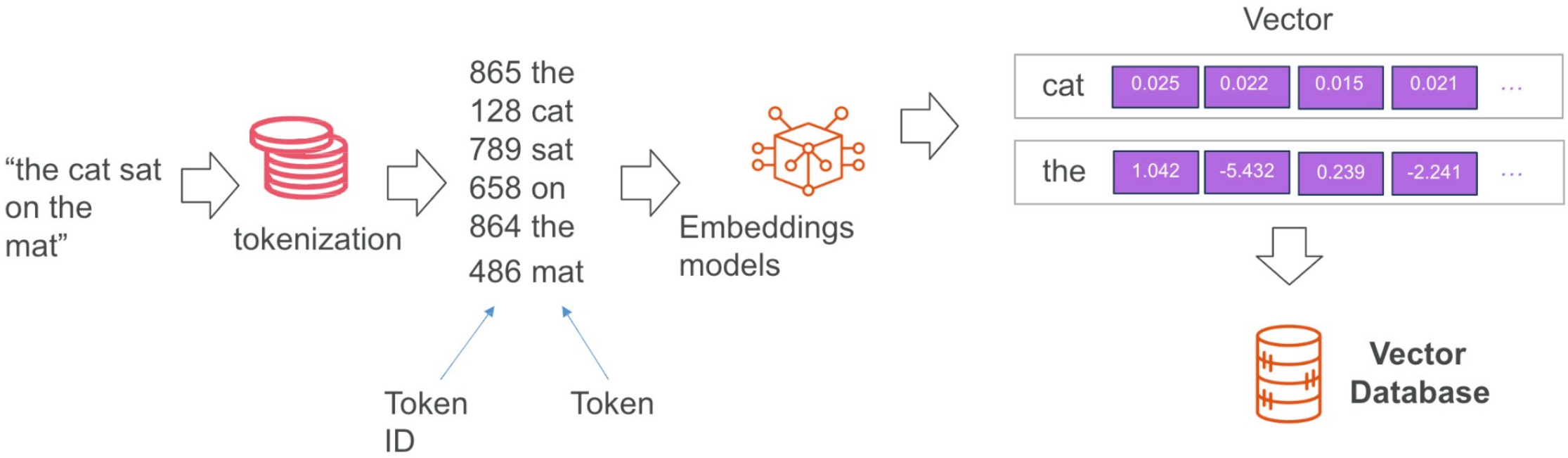
# How RAG Works




Retrieval finds the documents that resemble the question · Augmentation adds them to the prompt · Generation produces an answer grounded in what was retrieved.


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# Tokenization and Text Embedding

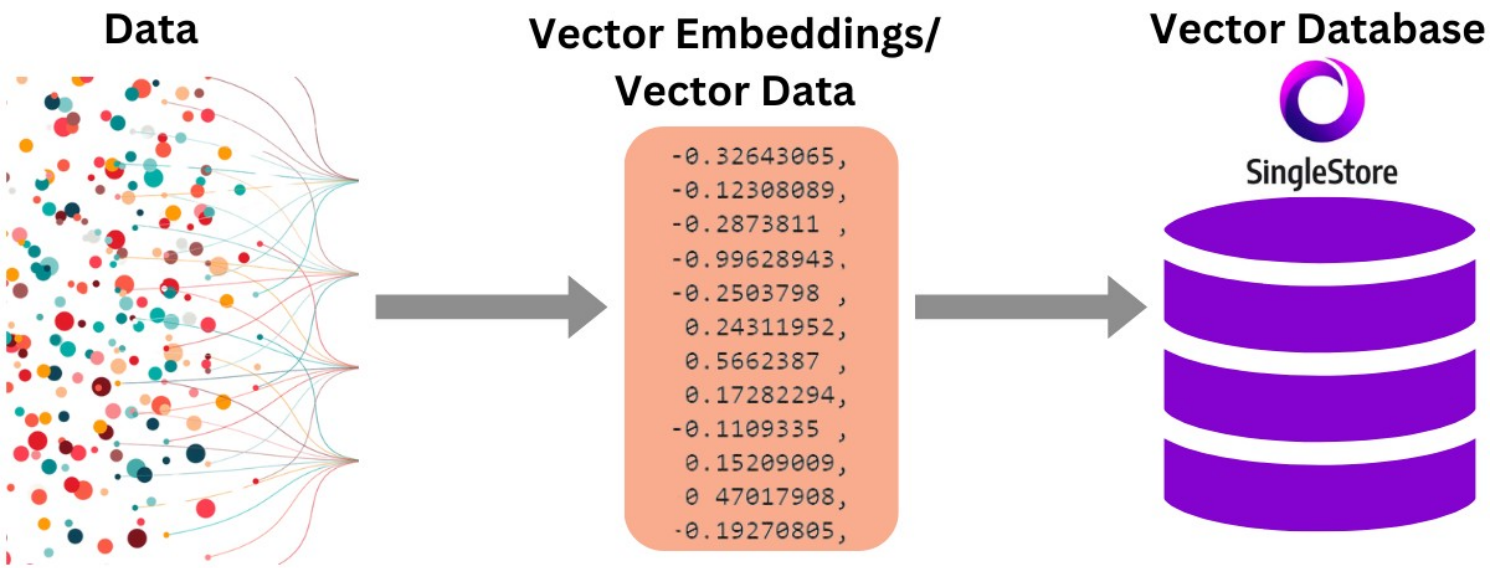



Text is split into tokens, then an embedding model turns it into a vector — a list of numbers that places similar meanings close together.


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TOPIC 03 · CONCEPT

# Vector Databases




A vector store holds the embeddings and finds the nearest neighbours to a question, fast.


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THE DESIGN QUESTION

# Why Not Just Paste Everything Into the Prompt?

- For twenty short documents, pasting them all would actually work.
- Add two hundred more and the prompt exceeds the model's context window.
- Long prompts also degrade quality — models attend less well to the middle of a very long context.
- And you pay for every token, on every single question, whether it was relevant or not.
- RAG turns the problem around: retrieve the few documents that resemble the question, and send only those.


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INGESTION THEN RETRIEVAL

# The RAG Pipeline

1. Collect the source documents
2. Split them into chunks
3. Embed each chunk into a vector
4. Store the vectors in a vector store
5. Embed the question and retrieve the nearest chunks
6. The LLM answers from what was retrieved


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WHERE RAG PROJECTS FAIL

# Chunking — the Biggest Lever on RAG Quality

1

## The rule

The right chunk is the smallest piece of text that still answers a question on its own.

2

## Too small

A brochure sliced into thirds separates the course name from its fee — retrieval finds one without the other.

3

## Too large

Irrelevant text crowds the prompt and dilutes what actually matters.

4

## Overlap

A small overlap between chunks stops a sentence being cut in half at a boundary.

5

## In this lab

Chunk size 4000, overlap 0 — one brochure becomes exactly one vector.

6

## The lesson

Most RAG projects fail quietly at chunking, not at the model.


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IN-MEMORY VS HOSTED

# Choosing a Vector Store

## Activity 5 uses this

- Simple Vector Store (built in)
  - No account, key or index needed
  - Running RAG in ten minutes
  - WIPED when n8n restarts
  - Good for: learning and prototypes

## What production needs

- A real vector database
  - Pinecone, Supabase (pgvector), Qdrant
  - Account, index and dimension to set up
  - Persists and scales; shared across instances
  - Good for: anything in production


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# The Rules That Keep RAG Working

- The ingestion and retrieval embedding models must be IDENTICAL — the numbers mean different things otherwise.
- The vector store's dimension must match the embedding model's output (Gemini embedding-001 = 3072).
- The memory key or index name must match between ingestion and retrieval.
- Turn Clear Store on when re-ingesting, or every document is stored — and retrieved — twice.
- topK controls how many chunks are retrieved: too few misses the answer, too many dilutes the prompt.


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THE INSTRUCTIONS THAT MATTER

# Grounding an Agent Against Hallucination

1

## Always call the tool first

Instruct the agent to search before answering any factual question.

2

## Answer only from what came back

Every factual claim must trace to a retrieved document.

3

## Give it permission to fail

'Say I don't have that in our course information' is a better answer than a confident invention.

4

## Name what it must never invent

A fee, a date, a duration, a course code, an instructor name, a discount.

5

## Hide the machinery

Never mention the vector store or that it is searching documents — the customer does not care.

6

## Test the traps

Ask about a course you do not run, a person who does not exist, and a discount never offered.


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WHAT YOU'LL DO

# Hands-On Labs — Retrieval-Augmented Generation (RAG)

1

**Lab 5 — RAG Customer Care Chatbot (Cook & Bake Academy)**

Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond


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TOPIC 03 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot (Cook & Bake Academy)

**LAB 5**

Cook & Bake Academy runs 20 courses across two campuses, and its two-person customer care team retypes the same answers all day. Every answer is already written in the course brochures. You build the full RAG loop: an ingestion workflow that embeds all 20 brochures into a vector store, and a chatbot that retrieves only the brochures resembling each question and answers strictly from those.

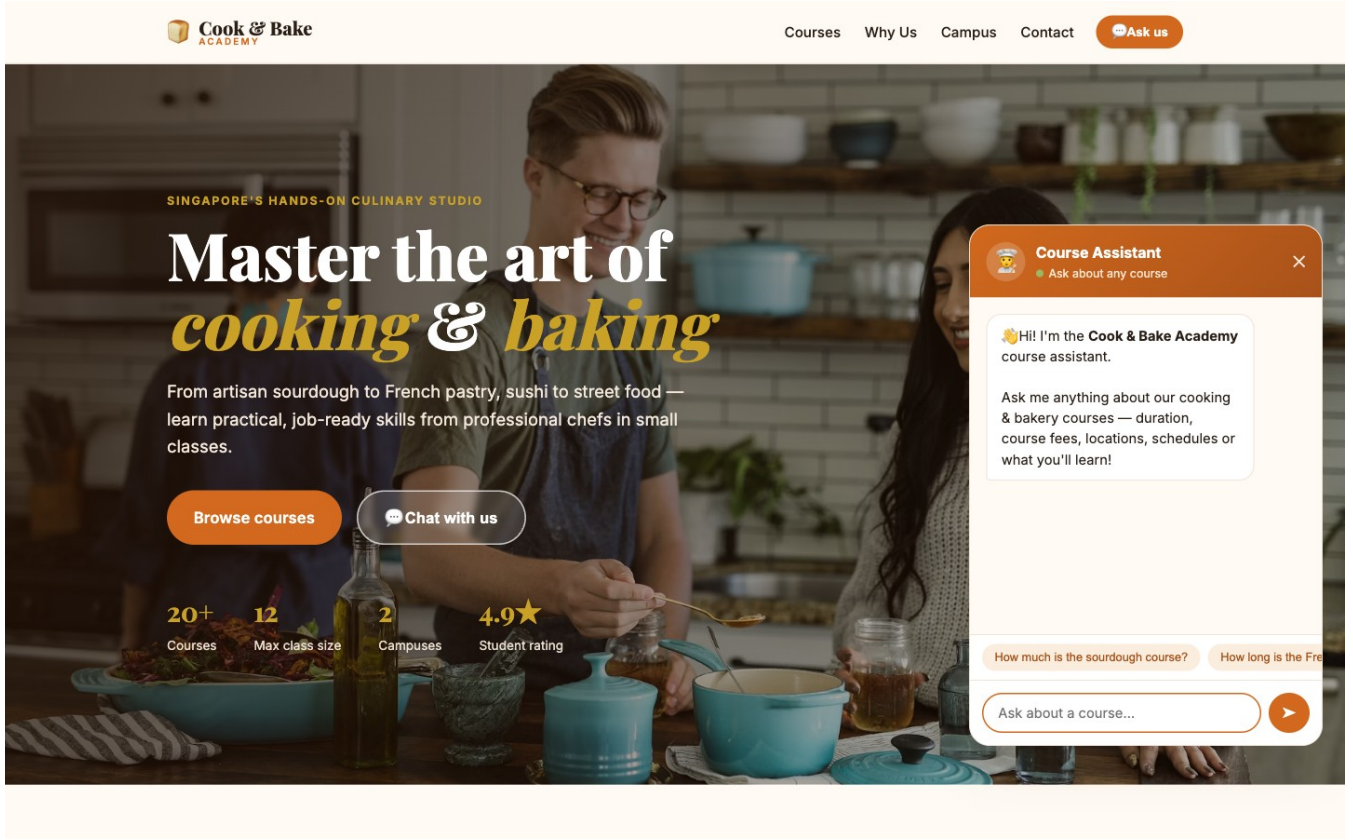
**You'll build**

**Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond**

**Tools:** n8n Simple Vector Store, Google Gemini Embeddings, Google Drive, AI Agent, Webhook


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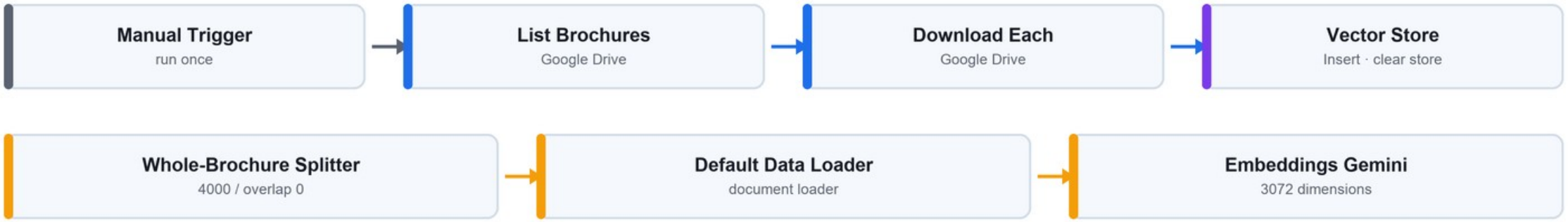
# Lab 5 — RAG Customer Care Chatbot — What You'll Build




The Cook & Bake Academy site with the course assistant open. Every answer it gives is grounded in the 20 brochures you ingested into the vector store.


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# Lab 5 — RAG Customer Care Chatbot — Ingestion Workflow




Ingestion — run once by hand. Twenty brochures are listed, downloaded, split (one brochure = one chunk), embedded into 3072-dimensional vectors and inserted into the store.


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LAB 5 · THE WORKFLOW

# Lab 5 — RAG Customer Care Chatbot — Answering Workflow




Answering — runs on every question. The agent embeds the question with the SAME model, retrieves the nearest brochures through the Query Data Tool, and answers only from those.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 1 OF 15

1

In Google Drive create a folder named 'brochures' and upload all 20 .txt files from the lab's brochures/ folder.

**WHY THIS STEP**

That folder becomes the academy's single source of truth. When a fee changes, it changes here and nowhere else.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 2 OF 15

2

Open one brochure and read it — everything the chatbot will ever say comes from files like this one.

**WHY THIS STEP**

If a fact is not in these files, a correctly-grounded chatbot cannot say it. That is the point of RAG.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 3 OF 15

3

Import lab5-Ingest-Simple-Vector-Store.json — this is the ingestion workflow you run once.

**WHY THIS STEP**

Ingestion and answering are separate workflows: you ingest once, then answer thousands of questions.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 4 OF 15

4

Open List Brochures in Folder, select YOUR brochures folder, and set your Google Drive credential.

**WHY THIS STEP**

Point it at your own folder, or you will be answering questions from the author's copy of the documents.


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# Lab 5 — RAG Customer Care Chatbot

STEP 5 OF 15

5

Open Simple Vector Store (Insert) and change the Memory Key from cookbake_brochures to cookbake_<your-name>, because the whole class shares one namespace.

**WHY THIS STEP**

The memory key is a global namespace on a shared instance. Leave the default and you will be answering questions from each other's brochures.


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# Lab 5 — RAG Customer Care Chatbot

STEP 6 OF 15

6

Confirm Clear Store is ON so that re-running replaces the 20 brochures instead of storing them twice.

**WHY THIS STEP**

Turn it off and run twice, and every search returns each brochure twice.


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# Lab 5 — RAG Customer Care Chatbot

STEP 7 OF 15

7

Open Whole-Brochure Splitter and note chunk size 4000 with overlap 0 — one brochure becomes exactly one vector, so the fee is never separated from the course name.

**WHY THIS STEP**

The default 1000-character splitter would slice a brochure into three: the fragment mentioning sourdough would not be the fragment with the price.


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# Lab 5 — RAG Customer Care Chatbot

STEP 8 OF 15

8

Open Embeddings Google Gemini, select your credential, and note the model gemini-embedding-001 produces 3072-dimensional vectors.

**WHY THIS STEP**

Remember this number. The store's dimension must equal the embedding model's output, and a mismatch is the top cause of failed inserts.


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# Lab 5 — RAG Customer Care Chatbot

STEP 9 OF 15

9

Click Execute workflow. Twenty files are listed, downloaded, embedded and stored in about thirty seconds.

**WHY THIS STEP**

Confirm exactly twenty vectors were inserted. More than twenty means Clear Store was off and you have duplicates.


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# Lab 5 — RAG Customer Care Chatbot

STEP 10 OF 15

10

Import LU5-rag-chatbot.json — the answering agent.

**WHY THIS STEP**

This is the half that runs on every question. It never sees nineteen of the twenty brochures.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 11 OF 15

11

Open Query Data Tool and set its Memory Key to the SAME cookbake_<your-name> you used during ingestion. A mismatch here is the most common failure in this lab.

**WHY THIS STEP**

If the keys differ, the agent searches an empty store and answers 'I don't have that' to everything — with no error anywhere.


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# Lab 5 — RAG Customer Care Chatbot

STEP 12 OF 15

12

Confirm the retrieval Embeddings node uses the identical model to ingestion — a question embedded by one model cannot be compared against brochures embedded by another.

**WHY THIS STEP**

The numbers mean different things between models. The comparison silently returns nonsense rather than failing.


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# Lab 5 — RAG Customer Care Chatbot

STEP 13 OF 15

13

Open AI Agent and read the grounding rules: always call the tool first, never invent a fee or an instructor, and say 'I don't have that in our course information' when the brochures are silent.

**WHY THIS STEP**

An ungrounded model asked for a price will produce one. It will be plausible, it will be wrong, and the customer will quote it back to you.


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LAB 5 · HANDS-ON

# Lab 5 — RAG Customer Care Chatbot

STEP 14 OF 15

14

Save, Activate, copy the Production URL, then serve the website folder and paste the URL into Lab configuration.

```
$ python3 -m http.server 8000
```

**WHY THIS STEP**

If the webhook is unset the page silently falls back to a local keyword search and looks like it works — watch for the yellow warning banner.


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# Lab 5 — RAG Customer Care Chatbot

STEP 15 OF 15

15

Click the chat button and ask 'How much is the sourdough course?', then work through all ten rows of sample-questions.csv.

**WHY THIS STEP**

TC7, TC8 and TC9 each plant something false and invite the model to agree. Any confident answer is a failure, however fluent.


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LAB 5 · VERIFY

# Lab 5 — RAG Customer Care Chatbot

## ✅ Test it

TC1 must quote BAK-101's exact fee from the brochure. Then run the three hallucination probes: TC7 (a course that does not exist), TC8 (an instructor name no brochure contains) and TC9 (a discount that was never offered). Any confident answer to these is a failure, however fluent.


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LAB DEBRIEF

# What You Learned — Lab 5

1. **What RAG does**
   RAG retrieves the few documents that resemble the question instead of pasting everything into the prompt.

2. **Embeddings**
   An embedding places similar meanings close together, so retrieval works even when no words match.

3. **Chunking**
   Chunking is the single biggest lever on RAG quality — the right chunk answers a question on its own.

4. **Match model and key**
   The ingestion and retrieval embedding models must be identical, and the memory keys must match.

5. **In-memory is temporary**
   n8n's Simple Vector Store is wiped when n8n restarts — fine for learning, unacceptable in production.


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TOPIC RECAP

# Recap — Retrieval-Augmented Generation (RAG)

1

## What RAG does

Lab 5 — you can now deploy a RAG application that grounds an agent in the organisation's own documents.

2

## Embeddings

An embedding places similar meanings close together, so retrieval works even when no words match.

3

## Chunking

Chunking is the single biggest lever on RAG quality — the right chunk answers a question on its own.


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WRAP-UP

# Course Summary & Next Steps


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LEARNING OUTCOMES

# What You Achieved

1

## LO1 · n8n and AI agents

Built a trigger-and-action workflow and explained how an LLM agent differs from generative AI.

2

## LO2 · Webhook-triggered agents

Put an AI agent behind a real website, with tools, structured output and an audit log.

3

## LO3 · Agentic chatbots

Designed a chatbot that collects contact details and refuses to give regulated advice.

4

## LO4 · Human in the loop

Paused a workflow on an approval gate so no drafted communication was sent unsupervised.

5

## LO5 · RAG

Ingested 20 documents into a vector store and grounded a chatbot so it stopped inventing answers.


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NEXT STEPS

# Taking This Back to Work

1

## Start with one painful process

Pick something repetitive, high-volume and low-risk. Automate that first.

2

## Map it before you build it

Draw the trigger, the decisions and the actions on paper. The canvas comes second.

3

## Decide where the human belongs

Anything irreversible, regulated or expensive keeps an approval gate.

4

## Ground the agent in your own data

RAG over your policies and documents beats a general model every time.

5

## Log everything from day one

You will need the audit trail long before you think you will.

6

## Iterate with real cases

Test with the questions you fear, not the ones you hope for.


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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

# Support

1

**Email**

enquiry@tertiaryinfotech.com

2

**Telephone**

+65 6100 0613

3

**Website**

www.tertiarycourses.com.sg

4

**LMS / TMS portal**

lms-tms.tertiaryinfotech.com


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# Assessment

1

## Written Assessment (SAQ)

1 hour. Short-answer questions covering the course knowledge.

2

## Practical Performance (PP)

1 hour. Hands-on n8n build tasks drawn from the five labs.

3

## Open book

Slides, Learner Guide and approved materials only.

4

## Submit on the LMS

Upload your completed answers at lms-tms.tertiaryinfotech.com.

5

## Digital attendance

Remember to take the Assessment digital attendance (TRAQOM).

6

## Appeals

An appeal process is available if required.


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ON ASSESSMENT DAY

# Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record


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# Digital Attendance (Mandatory)

1

## AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

## The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

## Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

## 75% minimum

At least 75% attendance is required to be eligible for assessment and funding.


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HAPPY AUTOMATING

# Thank You!

Keep your n8n instance running and keep building. The next process you automate is the one you were dreading this morning.


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